Econometrics: Advanced Methods

Short lecture notes for Research Metrics 2 (MQE, LMU Munich), Winter 2026/27

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Table of contents

Preface **4**
What the course covers . . . 4
Literature . . . 4
Software . . . 5
How to use the tutorials . . . 5
Conventions . . . 5

I Estimation principles **6**

1 Maximum Likelihood **7**
1.1 Parametric models . . . 7
1.2 The likelihood function and the MLE . . . 8
1.3 Score, Hessian and information . . . 9
1.4 Asymptotic properties of the MLE . . . 10
1.5 Estimating the variance matrix . . . 11
1.6 Misspecification and pseudo-ML . . . 11
1.7 Hypothesis testing: the trinity . . . 13
1.8 Computer tutorial 1 . . . 14
1.9 Exercises . . . 16

2 Method of Moments and GMM **17**
2.1 Moment conditions . . . 17
2.2 The method of moments estimator . . . 18
2.3 The generalized method of moments . . . 18
2.4 Asymptotic properties of GMM . . . 19
2.5 Efficient GMM . . . 19
2.6 Testing overidentifying restrictions . . . 20
2.7 Conditional moments and optimal instruments . . . 20
2.8 Linear instrumental variables as GMM . . . 21
2.9 Two-step estimators as joint GMM . . . 22
2.10 Further applications . . . 25
2.11 Computer tutorial 2 . . . 25
2.12 Exercises . . . 26

II Computation and flexible models **27**

3 Simulation-Based Estimation **28**

4 Nonparametric Methods **29**

5 Introduction to Statistical Learning **30**

References **31**

Appendices **33**

A Large-Sample Asymptotics **33**

B A Short R Primer **34**

Preface

These notes accompany the course *Econometrics: Advanced Methods (Research Metrics 2)* in the Munich Graduate School of Economics MQE programme. They are deliberately short. Each chapter states the key definitions and results, works through one or two examples in detail, and ends with a computer tutorial in R. Proofs are sketched where the argument is instructive and otherwise referenced.

The notes complement the lecture slides; they do not replace the textbooks listed below.

What the course covers

The objective is to introduce fundamental estimation methods in econometrics. We spend a fair amount of time on the theoretical properties of the estimators and also discuss practical aspects of implementation. The course is not a course on causality or identification, and it is not an R course, although both appear throughout.

Chapter	Topic	Computer tutorial
1	Maximum likelihood	CT 1: logit by hand, PML vs. ML
2	Method of moments and GMM, including IV	CT 2: GMM estimation of an ATE
3	Simulation-based estimation	CT 3: Monte Carlo integration, Halton draws, quadrature
4	Nonparametric density and regression estimation	CT 4: kernel density, local linear regression
5	Introduction to statistical learning	CT 5: ridge, lasso, elastic net
A	Large-sample asymptotics (assumed known; for reference)	
B	A short R primer	CT 0

Literature

- Hansen, B. E. (2022). *Econometrics*. Princeton University Press (B. E. Hansen 2022).
- Cameron, A. C. and Trivedi, P. K. (2005). *Microeconometrics: Methods and Applications*. Cambridge University Press (Cameron and Trivedi 2005).
- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. MIT Press (Wooldridge 2010).
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2023). *An Introduction to Statistical Learning*, 2nd ed. Springer (James et al. 2023).

- Casella, G. and Berger, R. L. (2002). *Statistical Inference*, 2nd ed. Duxbury (Casella and Berger 2002).

Chapter-specific readings are given at the start of each chapter.

Software

All tutorials use R. Install a recent version of R and an editor (RStudio or Positron), then install the packages used in the notes:

```
install.packages(c("sandwich", "lmtest", "MASS", "nleqslv",
                   "randtoolbox", "statmod", "glmnet", "haven"))
```

Every tutorial can be run from top to bottom. Data sets are in the `data/` folder of the course material on Moodle.

How to use the tutorials

Each tutorial first states the problems. Try them yourself before opening the solution boxes. Solutions are released after the tutorial session.

Conventions

Vectors and matrices are in bold, $\boldsymbol{\theta}$ denotes a generic parameter vector and $\boldsymbol{\theta}_0$ its true value. ln is the natural logarithm. $\xrightarrow{p}$ and $\xrightarrow{d}$ denote convergence in probability and in distribution.

Part I

Estimation principles

1 Maximum Likelihood

> **Readings**
>
> Wooldridge (2010), Ch. 13.1–13.7 and 13.11; Cameron and Trivedi (2005), Ch. 5.6–5.7; B. E. Hansen (2022), Ch. 10. Applications: Smith (2009) and Winkelmann (2004).

Maximum likelihood (ML) is the workhorse estimator for fully specified parametric models. This chapter introduces the likelihood principle, derives the large-sample properties of the MLE, discusses what happens when the assumed density is wrong, and closes with the classical trinity of tests.

1.1 Parametric models

We first consider unconditional distributions. A **parametric model** for a random variable Y is a complete probability function $f(y \mid \boldsymbol{\theta})$ that depends on an unknown parameter vector $\boldsymbol{\theta}$ in a **parameter space** Θ. We use the same notation for probability mass functions (discrete Y) and densities (continuous Y). For example, $Y \sim N(\mu, \sigma^2)$ has density $f(y \mid \mu, \sigma^2) = \sigma^{-1}\phi((y-\mu)/\sigma)$ with $\phi(z) = (2\pi)^{-1/2}\exp(-z^2/2)$, $\mu \in \mathbb{R}$ and $\sigma^2 > 0$. A parametric model need not be a textbook distribution; it can be built for a specific application.

Definition 1.1 (Model)**.** A model for a random sample is the assumption that $Y_1, \ldots, Y_n$ are i.i.d. with a known density (or mass) function $f(y \mid \boldsymbol{\theta})$ and unknown parameter $\boldsymbol{\theta} \in \Theta$.

Definition 1.2 (Correct specification)**.** A model is **correctly specified** if there exists a unique $\boldsymbol{\theta}_0 \in \Theta$ such that $f(y \mid \boldsymbol{\theta}_0) = f(y)$, the true density. $\boldsymbol{\theta}_0$ is the **true parameter value**. Uniqueness means that no other $\boldsymbol{\theta} \in \Theta$ satisfies $f(y \mid \boldsymbol{\theta}) = f(y \mid \boldsymbol{\theta}_0)$. A model is **misspecified** if no $\boldsymbol{\theta} \in \Theta$ satisfies $f(y \mid \boldsymbol{\theta}) = f(y)$.

Example 1.1 (Correct and incorrect models)**.** Let the true density be $f(y) = 2e^{-2y}$, $y > 0$.

- The exponential model $f(y \mid \lambda) = \lambda^{-1}e^{-y/\lambda}$ is correctly specified with $\lambda_0 = 1/2$.
- The gamma model $f(y \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1}e^{-\beta y}$ is correctly specified with $(\alpha_0, \beta_0) = (1, 2)$.
- The log-normal model $f(y \mid \theta, \nu) = (2\pi\nu)^{-1/2}y^{-1}\exp\{-(\ln y - \theta)^2/(2\nu)\}$ is misspecified: no (θ, ν) reproduces $2e^{-2y}$.

Uniqueness matters. A two-component normal mixture $p\,\phi_{\sigma_1}(y-\mu_1) + (1-p)\,\phi_{\sigma_2}(y-\mu_2)$ contains the standard normal $\phi(y)$, but at many parameter values: $(p, 0, 1, 0, 1)$ for any p, $(1, 0, 1, \mu_2, \sigma_2^2)$ for any (μ_2, σ_2^2), and so on. The model contains the truth, but $\boldsymbol{\theta}_0$ is not identified, so it is not correctly specified in the sense of Definition 1.2.

1.2 The likelihood function and the MLE

The **likelihood** is the joint density of the sample, evaluated at the observed data and viewed as a function of $\boldsymbol{\theta}$. Under independence it factorises, and under identical distributions all factors share the same form:

$$L_n(\boldsymbol{\theta}) = f(y_1, \dots, y_n \mid \boldsymbol{\theta}) = \prod_{i=1}^{n} f(y_i \mid \boldsymbol{\theta}), \qquad \ell_n(\boldsymbol{\theta}) \equiv \ln L_n(\boldsymbol{\theta}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{\theta}).$$

Definition 1.3 (Maximum likelihood estimator)**.** The MLE is the parameter value that maximises the likelihood,

$$\hat{\boldsymbol{\theta}}_n = \arg\max_{\boldsymbol{\theta} \in \Theta} L_n(\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta} \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{\theta}).$$

The second equality holds because $\ln(\cdot)$ is strictly increasing and scaling by $1/n$ does not change the maximiser.

The MLE is the parameter value under which the observed sample is most likely. Working with the average log-likelihood is convenient: it is a sample mean, so laws of large numbers and central limit theorems apply directly.

Example 1.2 (Bernoulli)**.** Let $Y_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\pi)$, so $f(y \mid \pi) = \pi^y (1-\pi)^{1-y}$ for $y \in \{0, 1\}$. Then

$$\ell_n(\pi) = \sum_{i=1}^{n} \left[y_i \ln \pi + (1 - y_i) \ln(1 - \pi) \right] = n \left[\bar{y} \ln \pi + (1 - \bar{y}) \ln(1 - \pi) \right].$$

The first-order condition $n\bar{y}/\pi - n(1-\bar{y})/(1-\pi) = 0$ gives $\hat{\pi} = \bar{y}$. Figure 1.1 plots the likelihood for two samples of size five.

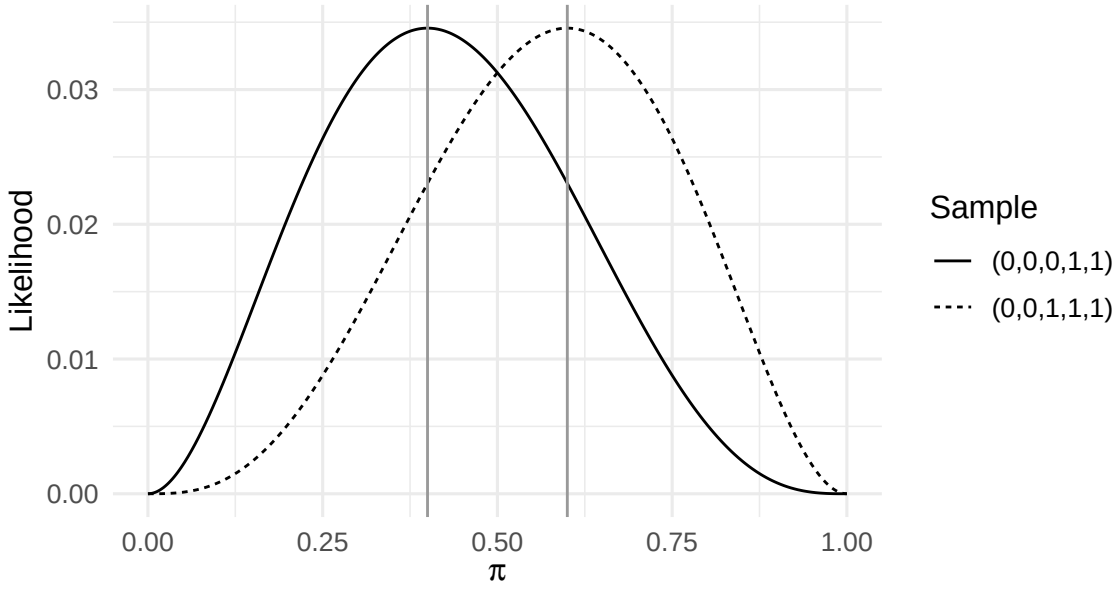


Figure 1.1: Bernoulli likelihood for the samples (0,0,0,1,1) and (0,0,1,1,1). The maximum is at the sample mean.

In practice we find the MLE in one of three ways:

1. **Solve the first-order condition** $\partial \ell_n(\boldsymbol{\theta})/\partial \boldsymbol{\theta} = \mathbf{0}$ and check the second-order condition. The FOC is necessary but not sufficient, and it only locates interior extrema; a maximum on the boundary of Θ need not satisfy it.
2. **Direct argument**, for instance by finding an upper bound of the likelihood that is attained at a unique point. The uniform distribution on $[0, \theta]$ is the standard case (Problem Set 1).
3. **Numerical optimisation** when no closed form exists (logit, probit, Tobit, ...). We then have to worry about convergence to a local rather than a global maximum.

Example 1.3 (Exponential)**.** Let $f(y \mid \lambda) = \lambda^{-1} e^{-y/\lambda}$, $y > 0$. The log-likelihood and its derivatives are

$$\ell_n(\lambda) = -n \ln \lambda - \frac{n\bar{y}}{\lambda}, \qquad \frac{d\ell_n}{d\lambda} = -\frac{n}{\lambda} + \frac{n\bar{y}}{\lambda^2}, \qquad \frac{d^2\ell_n}{d\lambda^2} = \frac{n}{\lambda^2} - \frac{2n\bar{y}}{\lambda^3}.$$

The unique root of the FOC is $\hat{\lambda} = \bar{y}$, and $d^2\ell_n(\hat{\lambda})/d\lambda^2 = -n/\bar{y}^2 < 0$ confirms a maximum.

Theorem 1.1 (Invariance)**.** *If $\hat{\boldsymbol{\theta}}$ is the MLE of $\boldsymbol{\theta} \in \mathbb{R}^m$, then for any function $\boldsymbol{\beta} = h(\boldsymbol{\theta}) \in \mathbb{R}^l$ the MLE of $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}} = h(\hat{\boldsymbol{\theta}})$.*

In Example 1.3, reparametrise with the rate $\beta = 1/\lambda$. Then $\ell_n(\beta) = n \ln \beta - \beta n \bar{y}$, maximised at $\hat{\beta} = 1/\bar{y} = h(\hat{\lambda})$, as the theorem predicts. Most estimators do not share this property; an unbiased estimator of λ, for example, does not deliver an unbiased estimator of $1/\lambda$.

1.3 Score, Hessian and information

Assume $f(y \mid \boldsymbol{\theta})$ is twice differentiable in $\boldsymbol{\theta} \in \mathbb{R}^k$.

Definition 1.4 (Score and Hessian)**.** The **score** is the $k \times 1$ gradient of the log-likelihood and the **Hessian** is the $k \times k$ matrix of second derivatives:

$$S_n(\boldsymbol{\theta}) = \frac{\partial \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \sum_{i=1}^{n} \underbrace{\frac{\partial \ln f(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}}_{S_i(\boldsymbol{\theta})}, \qquad H_n(\boldsymbol{\theta}) = \frac{\partial^2 \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'} = \sum_{i=1}^{n} \underbrace{\frac{\partial^2 \ln f(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'}}_{H_i(\boldsymbol{\theta})}.$$

The score measures how sensitive the log-likelihood is to the parameters; at an interior MLE, $S_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$. The Hessian is symmetric and measures curvature. If ℓ_n is strictly concave, H_n is negative definite and the maximiser is unique. Both are sums over observations because the log-likelihood is.

Theorem 1.2 (Properties of the likelihood)**.** *Suppose the model is correctly specified, the support of Y does not depend on $\boldsymbol{\theta}$, and $\boldsymbol{\theta}_0$ is in the interior of Θ. Then*

(i) $\mathbb{E}[S_i(\boldsymbol{\theta}_0)] = \mathbf{0}$*;*

(ii) ***information matrix equality****:* $\mathbb{E}[S_i(\boldsymbol{\theta}_0) S_i(\boldsymbol{\theta}_0)'] = -\mathbb{E}[H_i(\boldsymbol{\theta}_0)] \equiv \mathcal{I}_1(\boldsymbol{\theta}_0)$.

$\mathcal{I}_1$ is the (Fisher) information of a single observation and $\mathcal{I}(\boldsymbol{\theta}_0) = n\mathcal{I}_1(\boldsymbol{\theta}_0)$ that of the sample. Because the score has mean zero, the left side of (ii) is the variance of the score. Part (i) follows from differentiating $\int f(y \mid \boldsymbol{\theta})\, dy = 1$ under the integral sign; part (ii) from differentiating once more. The support condition guarantees that this exchange of differentiation and integration is allowed. The uniform $U[0, \theta]$ violates it, which is why its MLE behaves non-standardly.

Theorem 1.3 (Cramér–Rao lower bound)**.** *Under the same regularity conditions, any unbiased estimator $\tilde{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}_0$ satisfies* $\operatorname{Var}(\tilde{\boldsymbol{\theta}}) \succeq \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$.

Example 1.4 (Bernoulli, continued)**.** For one observation, $S_i(\pi) = \dfrac{y_i - \pi}{\pi(1-\pi)}$ and $H_i(\pi) = -\dfrac{y_i}{\pi^2} - \dfrac{1-y_i}{(1-\pi)^2}$. At π_0: $\mathbb{E}[S_i] = 0$, $\operatorname{Var}(S_i) = \dfrac{\pi_0(1-\pi_0)}{\pi_0^2(1-\pi_0)^2} = \dfrac{1}{\pi_0(1-\pi_0)}$, and $-\mathbb{E}[H_i] = \dfrac{1}{\pi_0} + \dfrac{1}{1-\pi_0} = \dfrac{1}{\pi_0(1-\pi_0)}$. The information equality holds and the CRLB is $\pi_0(1-\pi_0)/n$, which $\hat{\pi} = \bar{y}$ attains.

1.3.1 Conditional likelihood

Econometric models are usually about the distribution of y given covariates $\boldsymbol{x}$. Write the joint density as $f(y, \boldsymbol{x} \mid \boldsymbol{\delta}) = f(y \mid \boldsymbol{x}; \boldsymbol{\theta})\, f(\boldsymbol{x} \mid \boldsymbol{\gamma})$ with $\boldsymbol{\delta} = (\boldsymbol{\theta}', \boldsymbol{\gamma}')'$. If $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}$ share no elements and are not linked by restrictions,

$$\ell_n(\boldsymbol{\delta}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta}) + \sum_{i=1}^{n} \ln f(\boldsymbol{x}_i \mid \boldsymbol{\gamma}),$$

and $\boldsymbol{\theta}$ can be estimated from the **conditional log-likelihood** $\sum_i \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta})$ alone, leaving the distribution of $\boldsymbol{x}$ unspecified. All results above carry over with expectations taken over the joint distribution of $(y, \boldsymbol{x})$.

1.4 Asymptotic properties of the MLE

Theorem 1.4 (Consistency (Wooldridge 2010, Thm. 13.1))**.** *Suppose (a) the conditional density $f(y \mid \boldsymbol{x}; \boldsymbol{\theta})$ is correctly specified; (b) $\boldsymbol{\theta}_0$ is globally identified: for every $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$ in Θ, $P\{f(y \mid \boldsymbol{x}; \boldsymbol{\theta}) \neq f(y \mid \boldsymbol{x}; \boldsymbol{\theta}_0)\} > 0$; (c) Θ is compact and the log-likelihood is continuous in $\boldsymbol{\theta}$ with a dominance condition. Then $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$.*

The idea: $\frac{1}{n}\ell_n(\boldsymbol{\theta}) \xrightarrow{p} \mathbb{E}[\ln f(y \mid \boldsymbol{x}; \boldsymbol{\theta})]$ uniformly in $\boldsymbol{\theta}$, and the limit is uniquely maximised at $\boldsymbol{\theta}_0$ (this follows from the KLIC inequality in Section 1.6). The maximiser of the sample objective then converges to the maximiser of the limit.

Theorem 1.5 (Asymptotic normality and efficiency)**.** *Under additional regularity conditions (interior $\boldsymbol{\theta}_0$, twice continuously differentiable log-density, nonsingular $\mathcal{I}_1(\boldsymbol{\theta}_0)$),*

$$\sqrt{n}\,(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N\big(\mathbf{0},\, \mathcal{I}_1(\boldsymbol{\theta}_0)^{-1}\big).$$

The MLE is asymptotically efficient: its asymptotic variance attains the Cramér–Rao bound, and no consistent and asymptotically normal estimator has a smaller asymptotic variance under the same assumptions.

Sketch. A mean-value expansion of the FOC around $\boldsymbol{\theta}_0$ gives $\mathbf{0} = S_n(\hat{\boldsymbol{\theta}}) = S_n(\boldsymbol{\theta}_0) + H_n(\bar{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$, so

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \Big[-\tfrac{1}{n} H_n(\bar{\boldsymbol{\theta}})\Big]^{-1} \tfrac{1}{\sqrt{n}} S_n(\boldsymbol{\theta}_0).$$

By the LLN and consistency, $-\frac{1}{n}H_n(\bar{\boldsymbol{\theta}}) \xrightarrow{p} \mathcal{I}_1(\boldsymbol{\theta}_0) \equiv \boldsymbol{A}_0$. By the CLT and Theorem 1.2, $\frac{1}{\sqrt{n}} S_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{B}_0)$ with $\boldsymbol{B}_0 = \mathbb{E}[S_i S_i']$. Slutsky gives the general form $N(\mathbf{0}, \boldsymbol{A}_0^{-1}\boldsymbol{B}_0\boldsymbol{A}_0^{-1})$, which collapses to $\mathcal{I}_1^{-1}$ because $\boldsymbol{A}_0 = \boldsymbol{B}_0$ under correct specification. Keep this sandwich form in mind; it reappears in Section 1.6 and in GMM.

1.5 Estimating the variance matrix

In finite samples we use $\operatorname{Var}(\hat{\boldsymbol{\theta}}) \approx \frac{1}{n}\mathcal{I}_1(\boldsymbol{\theta}_0)^{-1} = \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ and replace $\mathcal{I}_1$ by a consistent estimator. Three options are available, plus a fourth that does not rely on the information equality.

Table 1.1: Variance estimators after ML

	Estimator of $\operatorname{Var}(\hat{\boldsymbol{\theta}})$	Requires
(i) Expected Hessian	$\Big[-\sum_i \mathbb{E}\big(H_i(\boldsymbol{\theta}) \mid \boldsymbol{x}_i\big)\Big]^{-1}_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$	closed form of $\mathbb{E}(H_i \mid \boldsymbol{x}_i)$
(ii) Observed Hessian	$\Big[-H_n(\hat{\boldsymbol{\theta}})\Big]^{-1}$	second derivatives (analytic or numerical)
(iii) Outer product of gradients (OPG, BHHH)	$\Big[\sum_i S_i(\hat{\boldsymbol{\theta}}) S_i(\hat{\boldsymbol{\theta}})'\Big]^{-1}$	first derivatives only
(iv) Robust (sandwich)	$\big[-H_n(\hat{\boldsymbol{\theta}})\big]^{-1}\Big[\sum_i S_i S_i'\Big]\big[-H_n(\hat{\boldsymbol{\theta}})\big]^{-1}$	both

(i)–(iii) are all consistent under correct specification and typically differ in finite samples. The observed Hessian (ii) is the default in most software. The OPG (iii) is the cheapest but has the poorest finite-sample behaviour. The sandwich (iv) remains valid when the density is misspecified, as we see next.

For the Bernoulli model, all three of (i)–(iii) coincide at $\hat{\pi} = \bar{y}$ and equal $\bar{y}(1-\bar{y})/n$. Verifying this is a short exercise: use $\sum_i (y_i - \bar{y})^2 = n\bar{y}(1-\bar{y})$ for binary y_i.

1.5.1 Example: the normal linear model

Let $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \mid \boldsymbol{x}_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Then

$$\ell_n(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2.$$

Setting the derivatives to zero yields

$$\hat{\boldsymbol{\beta}}_{ML} = \Big(\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\Big)^{-1}\sum_i \boldsymbol{x}_i y_i, \qquad \hat{\sigma}^2_{ML} = \frac{1}{n}\sum_i (y_i - \boldsymbol{x}_i'\hat{\boldsymbol{\beta}})^2.$$

$\hat{\boldsymbol{\beta}}_{ML}$ is OLS. $\hat{\sigma}^2_{ML}$ lacks the degrees-of-freedom correction, so it is biased downward in finite samples, but the bias vanishes as $n \to \infty$.

1.6 Misspecification and pseudo-ML

What does the MLE estimate when the model is wrong? The answer uses a measure of distance between densities.

Definition 1.5 (Kullback–Leibler divergence)**.** For densities f and g, $\operatorname{KLIC}(f, g) = \int f(y)\ln\dfrac{f(y)}{g(y)}\,dy$.

KLIC is not symmetric, $\mathrm{KLIC}(f,g) \neq \mathrm{KLIC}(g,f)$ in general, so it is not a distance in the metric sense. Its key properties are $\mathrm{KLIC}(f,f) = 0$, $\mathrm{KLIC}(f,g) \geq 0$ (by Jensen's inequality) and therefore $f = \arg\min_g \mathrm{KLIC}(f,g)$. Applied to a parametric family $f_{\boldsymbol{\theta}} = f(\cdot \mid \boldsymbol{\theta})$: if the model is correct, $\boldsymbol{\theta}_0 = \arg\min_{\boldsymbol{\theta}} \mathrm{KLIC}(f, f_{\boldsymbol{\theta}})$.

When the model is misspecified, the parametric family may still be a good **approximating model**, for example a log-normal density for wages. The natural target is then the best-fitting member of the family.

Definition 1.6 (Pseudo-true parameter)**.** $\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\Theta} \mathrm{KLIC}(f, f_{\boldsymbol{\theta}})$.

Since $\mathrm{KLIC}(f, f_{\boldsymbol{\theta}}) = \int f \ln f \, dy - \mathbb{E}[\ln f(y \mid \boldsymbol{\theta})]$ and only the second term depends on $\boldsymbol{\theta}$, $\boldsymbol{\theta}^*$ maximises $\mathbb{E}[\ln f(y \mid \boldsymbol{\theta})]$. The MLE maximises the sample analogue of this expectation. ML therefore has two roles: it estimates $\boldsymbol{\theta}_0$ when the model is correct and $\boldsymbol{\theta}^*$ when it is not. In the second case we call it the **pseudo-** or **quasi-MLE** (PMLE).

Theorem 1.6 (Asymptotic distribution of the PMLE (Huber 1967; White 1982))**.** *Under regularity conditions,* $\hat{\boldsymbol{\theta}}_{PMLE} \xrightarrow{p} \boldsymbol{\theta}^*$ *and*

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{PMLE} - \boldsymbol{\theta}^*) \xrightarrow{d} N(\mathbf{0},\ \boldsymbol{A}^{*-1}\boldsymbol{B}^*\boldsymbol{A}^{*-1}), \qquad \boldsymbol{A}^* = -\mathbb{E}[H_i(\boldsymbol{\theta}^*)], \quad \boldsymbol{B}^* = \mathbb{E}[S_i(\boldsymbol{\theta}^*)S_i(\boldsymbol{\theta}^*)'].$$

The information equality fails, so $\boldsymbol{A}^* \neq \boldsymbol{B}^*$ and only the sandwich estimator (iv) in Table 1.1 is consistent. In general, $\boldsymbol{\theta}^* \neq \boldsymbol{\theta}_0$ and the PMLE is inconsistent for the parameter we care about. There is one important exception.

1.6.1 The linear exponential family

A density belongs to the **linear exponential family** (LEF) if it can be written

$$f(y \mid \mu) = \exp\{a(\mu) + b(y) + c(\mu)\, y\}, \qquad \mathbb{E}[y] = \mu, \quad \mathrm{Var}(y) = [c'(\mu)]^{-1}.$$

Table 1.2: Leading members of the linear exponential family

Distribution	$a(\mu)$, $b(y)$, $c(\mu)$	$\mathbb{E}[y]$	$\mathrm{Var}(y)$
Normal (σ^2 known)	$-\frac{\mu^2}{2\sigma^2} - \frac{1}{2}\ln(2\pi\sigma^2)$, $-\frac{y^2}{2\sigma^2}$, $\frac{\mu}{\sigma^2}$	μ	σ^2
Bernoulli	$\ln(1-\mu)$, 0, $\ln\frac{\mu}{1-\mu}$	μ	$\mu(1-\mu)$
Exponential	$-\ln\mu$, 0, $-1/\mu$	μ	μ^2
Poisson	$-\mu$, $-\ln y!$, $\ln\mu$	μ	μ

Theorem 1.7 (Consistency of LEF-based PMLE (Gouriéroux, Monfort, and Trognon 1984))**.** *If the assumed density is LEF and the conditional mean* $\mathbb{E}[y \mid \boldsymbol{x}] = m(\boldsymbol{x}, \boldsymbol{\theta}_0)$ *is correctly specified, the PMLE of* $\boldsymbol{\theta}$ *is consistent, whatever the true distribution of* y *given* $\boldsymbol{x}$*.*

The reason is that the LEF score is proportional to $y - m(\boldsymbol{x}, \boldsymbol{\theta})$, so its expectation is zero at $\boldsymbol{\theta}_0$ as soon as the mean is right. The practical consequence is important: Poisson regression consistently estimates an exponential conditional mean for *any* nonnegative outcome, count or not, provided inference uses the sandwich variance. Computer tutorial 1 illustrates this with overdispersed count data.

1.7 Hypothesis testing: the trinity

We test $H_0 : \boldsymbol{h}(\boldsymbol{\theta}_0) = \mathbf{0}$ against $H_1 : \boldsymbol{h}(\boldsymbol{\theta}_0) \neq \mathbf{0}$, where $\boldsymbol{h} : \mathbb{R}^k \to \mathbb{R}^q$ imposes $q \leq k$ restrictions with full-rank Jacobian $\boldsymbol{R}(\boldsymbol{\theta}) = \partial \boldsymbol{h}(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}'$. Linear restrictions are the special case $\boldsymbol{h}(\boldsymbol{\theta}) = \boldsymbol{R}\boldsymbol{\theta} - \boldsymbol{r}$. For example, with $k = 3$, $H_0 : \theta_1 = 1,\ \theta_2 - \theta_3 = 2$ has $q = 2$ and

$$\boldsymbol{R} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \boldsymbol{r} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Let $\hat{\boldsymbol{\theta}}_U$ be the unrestricted MLE and $\hat{\boldsymbol{\theta}}_R$ the MLE subject to $\boldsymbol{h}(\boldsymbol{\theta}) = \mathbf{0}$. A restriction such as $\beta_1 = c\beta_2$ can often be imposed by reparametrisation: $x_1\beta_1 + x_2\beta_2 = (c\,x_1 + x_2)\beta_2$, so the restricted model has a single regressor.

Theorem 1.8 (LR, Wald and LM tests)**.** *Under H_0 and correct specification, each of the following converges in distribution to χ^2_q:*

$$\begin{aligned} LR &= 2\big[\ell_n(\hat{\boldsymbol{\theta}}_U) - \ell_n(\hat{\boldsymbol{\theta}}_R)\big], \\ W &= \boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)' \Big[\boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)\, \widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}_U)\, \boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)'\Big]^{-1} \boldsymbol{h}(\hat{\boldsymbol{\theta}}_U), \\ LM &= S_n(\hat{\boldsymbol{\theta}}_R)'\, \widehat{\mathrm{Var}}(S_n(\hat{\boldsymbol{\theta}}_R))^{-1} S_n(\hat{\boldsymbol{\theta}}_R), \qquad \widehat{\mathrm{Var}}(S_n(\hat{\boldsymbol{\theta}}_R)) = \mathcal{I}(\hat{\boldsymbol{\theta}}_R) \text{ or } -H_n(\hat{\boldsymbol{\theta}}_R). \end{aligned}$$

The three statistics are asymptotically equivalent under H_0 and under local alternatives.

The intuition differs (Figure 1.2):

- **LR**: if the restriction is valid, imposing it should not reduce the maximised log-likelihood by much.
- **Wald**: if the restriction is valid, $\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)$ should be close to zero, because $\hat{\boldsymbol{\theta}}_U$ is consistent.
- **LM (score)**: if the restriction is valid, the slope of the log-likelihood at $\hat{\boldsymbol{\theta}}_R$ should be close to zero, because $\hat{\boldsymbol{\theta}}_R$ is then close to the unrestricted maximum.

For a single coefficient, $H_0 : \theta_j = 0$, the Wald statistic is the square of the usual t-ratio $\hat{\theta}_j / \mathrm{se}(\hat{\theta}_j)$.

The three statistics for this sample:

```
# y: 50 draws from an exponential with lambda = 1.5 (set.seed(42)); H0: lambda
↪  = 1.2
info <- function(l) n / l^2                  # Fisher information of the sample
LR <- 2 * (ll(l_hat) - ll(l0))
W  <- (l_hat - l0)^2 * info(l_hat)           # uses Var(l_hat) = l_hat^2 / n
LM <- slope^2 / info(l0)                     # score and information at l0
round(c(LR = LR, Wald = W, LM = LM, crit_5pct = qchisq(0.95, 1)), 3)
```

```
#>        LR      Wald        LM crit_5pct
#>     6.907     4.361     8.782     3.841
```

They differ in finite samples but lead to the same decision here. Which to use in practice?

1. **Computation.** LR needs both models, Wald only the unrestricted one, LM only the restricted one. LM is attractive when the restricted model is much simpler, e.g. when the restriction makes a nonlinear model linear.

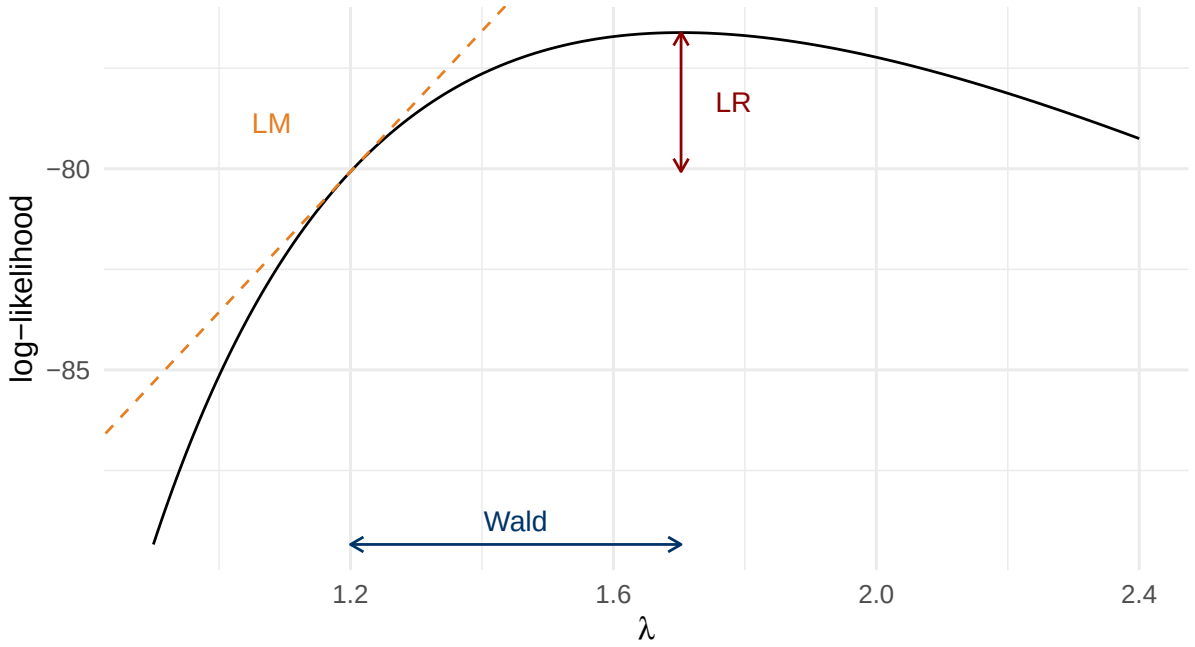


Figure 1.2: The trinity for the exponential model, testing $H_0 : \lambda = 1.2$ with n = 50 simulated observations. LR compares heights, Wald compares horizontal distance, LM compares the slope at the restricted value.

2. **Invariance.** Only LR is invariant to how the model and the restriction are parametrised. The Wald statistic for $H_0 : \theta = 1$ and for the equivalent $H_0 : 1/\theta = 1$ can differ substantially in small samples (Cameron and Trivedi 2005, sec. 7.2).
3. **Robustness.** Under misspecification, LR and the information-based LM are no longer χ^2. The Wald test with the sandwich variance remains valid, which is why robust Wald tests dominate applied work.

1.8 Computer tutorial 1

Learning goals. Write a log-likelihood as an R function and maximise it with `optim`; compute Hessian, OPG and sandwich variance estimators by hand and check them against `glm` and `sandwich`; see in a Monte Carlo study why pseudo-ML needs robust standard errors.

Packages. `sandwich`, `lmtest`, `MASS`. **Data.** `mroz.csv` (Mroz 1987), 753 married women in 1975.

1.8.1 Problem 1: Logit model for labour force participation

Variable	Description
`inlf`	= 1 if in labour force, 1975
`hours`	hours worked, 1975
`kidslt6`	number of kids under 6
`kidsge6`	number of kids aged 6–18
`age`	woman's age in years
`educ`	years of schooling

Variable	Description
`wage`	estimated hourly wage from earnings (0 if not working)
`faminc`	family income, 1975
`exper`	actual labour market experience
`expersq`	experience squared

a) Create non-wife income in thousand dollars, `nwifeinc` = (`faminc` − `wage` × `hours`)/1000.

b) Estimate the linear probability model (LPM)

$$\texttt{inlf} = \beta_0 + \beta_1 \texttt{nwifeinc} + \beta_2 \texttt{educ} + \beta_3 \texttt{exper} + \beta_4 \texttt{expersq} + \beta_5 \texttt{age} + \beta_6 \texttt{kidslt6} + \beta_7 \texttt{kidsge6} + \varepsilon.$$

Why are heteroskedasticity-robust standard errors required here? How many fitted values lie outside $[0, 1]$?

c) Write down the logit model and its log-likelihood formally. Derive the score and the Hessian of observation i.

d) Estimate the logit model with the regressors from (b) in two ways.

1. Code the **negative** log-likelihood as an R function `negll(b, y, X)` and minimise it with `optim(..., method = "BFGS", hessian = TRUE)`, using the LPM coefficients as starting values. Supply the analytical gradient as well.
2. Use `glm(..., family = binomial(link = "logit"))`.

e) Compute the variance matrix of $\hat{\boldsymbol{\beta}}$ in four ways: (i) from the Hessian returned by `optim`; (ii) the inverse of the negative analytical Hessian; (iii) the inverse of the OPG; (iv) the sandwich. Compare with `vcov()` and `sandwich()` applied to the `glm` object.

f) Compute the average partial effects (APEs) of `educ` and `kidslt6` from the logit model and compare them with the LPM coefficients.

1.8.2 Problem 2: PML vs. ML, a Monte Carlo study

The goal is to see how ML behaves when the likelihood is misspecified but the conditional mean is right (see "Misspecification and pseudo-ML" in Chapter 1 of the lecture notes).

Correct density.

a) Simulate $n = 1000$ observations from $y_i \mid \boldsymbol{x}_i \sim \text{Poisson}(\lambda_i)$ with $\lambda_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})$, $(\beta_0, \beta_1, \beta_2) = (0.2, 0.1, 0)$ and x_1, x_2 independent $N(0, 1)$.

b) Estimate $\boldsymbol{\beta}$ with `glm(..., family = poisson)`.

c) Using the score and Hessian of the Poisson model from Problem Set 1, Problem 3, compute Hessian-based and sandwich standard errors by hand. Check them against `glm` and `coeftest(..., vcov = sandwich)`.

Misspecified density.

d) Now draw y_i from a negative binomial distribution with the same mean $\mu_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})$ and variance $\text{Var}(y_i \mid \boldsymbol{x}_i) = \mu_i + \mu_i^2/\theta$ with $\theta = 3$.

e) Estimate the mean function by (i) Poisson PML with Hessian-based variance, (ii) Poisson PML with sandwich variance, (iii) negative binomial ML (`MASS::glm.nb`) with Hessian-based variance.

f) Repeat (d)–(e) for 1000 Monte Carlo samples and fill in the table:

	Poisson PML (Hessian)	Poisson PML (sandwich)	NegBin ML (Hessian)
MC mean of $\hat{\beta}_2$			
MC mean of s.e.($\hat{\beta}_2$)			
MC std. dev. of $\hat{\beta}_2$			
Rejection rate of $H_0 : \beta_2 = 0$ at 5%			

g) Discuss your results.

1.9 Exercises

Problem Set 1 contains the analytical exercises for this chapter:

- **Problem 1**: a regular one-parameter model from start to finish (score, information, bias and the Cramér–Rao bound, asymptotics, invariance) and the three tests, including the non-invariance of the Wald test.
- **Problem 2**: the uniform distribution as a non-regular case, with an n-rate, non-normal limit.
- **Problem 3**: Poisson regression, its score and Hessian, and pseudo-ML under overdispersion. Computer tutorial 1, Problem 2 builds on it.
- **Problem 4**: canonical links in the linear exponential family.

2 Method of Moments and GMM

Readings

Cameron and Trivedi (2005), Ch. 6.1–6.5; B. E. Hansen (2022), Ch. 12 (instrumental variables) and Ch. 13 (GMM); Wooldridge (2010), Ch. 8 and 14. Potential outcomes and weighting (Section 2.9): Imbens and Wooldridge (2009), Sections 1–5; Wooldridge (2010), Ch. 21.1–21.3. Applications: Arellano and Bond (1991) and Imai and Ratkovic (2014).

Maximum likelihood requires the full distribution of the data. Many economic models deliver less: a set of expectations that should be zero at the true parameter value. The method of moments (MM) and its generalization, the generalized method of moments (GMM), turn such moment conditions directly into estimators. OLS, IV, 2SLS and ML are all special cases, and the GMM framework gives a single recipe for their asymptotic distribution and for efficient estimation when there are more conditions than parameters (L. P. Hansen 1982).

2.1 Moment conditions

Let $\boldsymbol{w}_i$ collect all observed variables for unit i (outcomes, regressors, instruments) and let $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^k$.

Definition 2.1 (Moment function)**.** A $q \times 1$ vector function $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})$ with $q \geq k$ is a **moment function** if

$$\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \mathbf{0}.$$

This equation is a (population) **moment condition** or moment restriction.

A moment function depends only on the data and the parameters, it has at least as many elements as there are parameters, and its expectation is zero at the true value. Three examples recur throughout the chapter.

Example 2.1 (Familiar estimators as moment estimators)**.**

1. **Mean.** $\psi(y_i, \mu) = y_i - \mu$. The sample analogue $\frac{1}{n}\sum_i (y_i - \hat{\mu}) = 0$ gives $\hat{\mu} = \bar{y}$.
2. **OLS.** In $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + u_i$, the assumption $\mathbb{E}[u_i \mid \boldsymbol{x}_i] = 0$ implies the k conditions $\mathbb{E}[\boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})] = \mathbf{0}$. Solving the sample analogue gives $\hat{\boldsymbol{\beta}} = \left(\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\right)^{-1}\sum_i \boldsymbol{x}_i y_i$.
3. **Maximum likelihood.** Under correct specification the score has mean zero (Theorem 1.2). With $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}) = \partial \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta})/\partial\boldsymbol{\theta}$, the MM estimator is the root of the score, i.e. the MLE.

2.2 The method of moments estimator

Definition 2.2 (Method of moments estimator)**.** If $q = k$, the MM estimator $\hat{\boldsymbol{\theta}}$ solves the k sample moment conditions

$$\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) \equiv \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{\psi}(\boldsymbol{w}_i, \hat{\boldsymbol{\theta}}) = \mathbf{0}.$$

MM estimators are consistent whenever a law of large numbers makes the sample moments converge to the population moments and $\boldsymbol{\theta}_0$ is the unique solution of $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})] = \mathbf{0}$. For OLS,

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \Big(\tfrac{1}{n}\textstyle\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\Big)^{-1}\tfrac{1}{n}\textstyle\sum_i \boldsymbol{x}_i u_i \;\xrightarrow{p}\; \boldsymbol{\beta} + \mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}\mathbb{E}[\boldsymbol{x}_i u_i] = \boldsymbol{\beta},$$

provided $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ is nonsingular.

Theorem 2.1 (Asymptotic distribution of the MM estimator)**.** *Under regularity conditions and i.i.d. sampling,*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0},\; \boldsymbol{A}_0^{-1}\boldsymbol{V}_0\boldsymbol{A}_0^{-1\prime}), \qquad \boldsymbol{A}_0 = \mathbb{E}\Big[\frac{\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}'}\Big], \quad \boldsymbol{V}_0 = \mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)'].$$

Sketch. A mean-value expansion of $\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$ around $\boldsymbol{\theta}_0$ gives $\mathbf{0} = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) + \boldsymbol{A}_n(\bar{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ with $\boldsymbol{A}_n = \frac{1}{n}\sum_i \partial\boldsymbol{\psi}_i/\partial\boldsymbol{\theta}'$. Hence $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\boldsymbol{A}_n^{-1}\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0)$. The LLN gives $\boldsymbol{A}_n \xrightarrow{p} \boldsymbol{A}_0$, the CLT gives $\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{V}_0)$, and Slutsky does the rest. This is the same argument as for the MLE in Chapter 1; the MLE simply has $\boldsymbol{A}_0 = -\mathcal{I}_1$ and $\boldsymbol{V}_0 = \mathcal{I}_1$.

The variance is estimated by the sandwich $\hat{\boldsymbol{A}}^{-1}\hat{\boldsymbol{V}}\hat{\boldsymbol{A}}^{-1\prime}$ with $\hat{\boldsymbol{A}} = \frac{1}{n}\sum_i \partial\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})/\partial\boldsymbol{\theta}'$ and $\hat{\boldsymbol{V}} = \frac{1}{n}\sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})'$.

Example 2.2 (OLS under heteroskedasticity)**.** With $\boldsymbol{\psi} = \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})$ we get $\boldsymbol{A}_0 = -\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ and $\boldsymbol{V}_0 = \mathbb{E}[u_i^2\boldsymbol{x}_i\boldsymbol{x}_i']$, so

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0},\; \mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}\mathbb{E}[u_i^2\boldsymbol{x}_i\boldsymbol{x}_i']\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}).$$

White's heteroskedasticity-robust variance estimator is the MM sandwich for OLS. Likewise, the PML sandwich in Section 1.6 is the MM sandwich with the score as moment function.

2.3 The generalized method of moments

When $q > k$ the model is **overidentified**: in general no $\boldsymbol{\theta}$ sets all q sample moments to zero at once. GMM instead makes them as small as possible in a quadratic metric.

Example 2.3 (Two sets of conditions for one regression)**.** If, in addition to $\mathbb{E}[u_i \mid \boldsymbol{x}_i] = 0$, the errors are conditionally symmetric, then $\mathbb{E}[u_i^3 \mid \boldsymbol{x}_i] = 0$ and

$$\mathbb{E}\begin{pmatrix} \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta}) \\ \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^3 \end{pmatrix} = \mathbf{0},$$

which gives $2k$ conditions for k parameters.

Definition 2.3 (GMM estimator)**.** Let $\boldsymbol{W}_n$ be a $q \times q$ symmetric positive definite weighting matrix, possibly data dependent. The GMM estimator is

$$\hat{\boldsymbol{\theta}}(\boldsymbol{W}_n) = \arg\min_{\boldsymbol{\theta} \in \Theta} \; Q_n(\boldsymbol{\theta}), \qquad Q_n(\boldsymbol{\theta}) = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta})' \, \boldsymbol{W}_n \, \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}).$$

The first-order conditions are

$$\boldsymbol{A}_n(\hat{\boldsymbol{\theta}})' \, \boldsymbol{W}_n \, \bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \boldsymbol{0}, \qquad \boldsymbol{A}_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \frac{\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'},$$

i.e. GMM sets k linear combinations of the q sample moments to zero. If $q = k$, $\boldsymbol{A}_n$ is square and (generically) invertible, so the FOC reduce to $\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \boldsymbol{0}$: in the just-identified case GMM is MM, whatever $\boldsymbol{W}_n$.

2.4 Asymptotic properties of GMM

Theorem 2.2 (Consistency and asymptotic normality of GMM)**.** *Assume i.i.d. data and*

(i) ***identification:*** $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})] = \boldsymbol{0}$ *if and only if* $\boldsymbol{\theta} = \boldsymbol{\theta}_0$*;*

(ii) $\boldsymbol{A}_0 = \mathbb{E}[\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)/\partial \boldsymbol{\theta}']$ *is finite with full column rank* k*;*

(iii) $\boldsymbol{W}_n \xrightarrow{p} \boldsymbol{W}_0$*, symmetric positive definite;*

(iv) $\sqrt{n}\, \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\boldsymbol{0}, \boldsymbol{V}_0)$*,* $\boldsymbol{V}_0 = \mathbb{E}[\boldsymbol{\psi}_i \boldsymbol{\psi}_i']$ *positive definite;*

plus standard regularity conditions (compact Θ*, uniform convergence, smoothness). Then* $\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$ *and*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N\Big(\boldsymbol{0}, \; (\boldsymbol{A}_0' \boldsymbol{W}_0 \boldsymbol{A}_0)^{-1} \boldsymbol{A}_0' \boldsymbol{W}_0 \boldsymbol{V}_0 \boldsymbol{W}_0 \boldsymbol{A}_0 (\boldsymbol{A}_0' \boldsymbol{W}_0 \boldsymbol{A}_0)^{-1}\Big).$$

The proof follows the MM sketch: expand the FOC around $\boldsymbol{\theta}_0$ and solve, $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \approx -(\boldsymbol{A}_0' \boldsymbol{W}_0 \boldsymbol{A}_0)^{-1} \boldsymbol{A}_0' \boldsymbol{W}_0 \sqrt{n}\, \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0)$. Two features are worth stressing. First, **the weighting matrix does not affect consistency**, only the asymptotic variance. Second, condition (ii) is the local version of identification: if $\boldsymbol{A}_0$ has deficient rank, the moments do not pin down all parameters, which is the essence of weak or irrelevant instruments.

The variance is estimated by replacing $\boldsymbol{A}_0, \boldsymbol{V}_0, \boldsymbol{W}_0$ by $\hat{\boldsymbol{A}} = \boldsymbol{A}_n(\hat{\boldsymbol{\theta}})$, $\hat{\boldsymbol{V}} = \frac{1}{n} \sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}}) \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})'$ and $\boldsymbol{W}_n$, and dividing by n.

2.5 Efficient GMM

Theorem 2.3 (Optimal weighting matrix)**.** *Among all choices of* $\boldsymbol{W}_0$*,* $\boldsymbol{W}_0^* = \boldsymbol{V}_0^{-1}$ *minimizes the asymptotic variance (in the matrix sense of Chapter 1). The minimum is*

$$\operatorname{Avar}\left(\hat{\boldsymbol{\theta}}(\boldsymbol{V}_0^{-1})\right) = \left(\boldsymbol{A}_0' \boldsymbol{V}_0^{-1} \boldsymbol{A}_0\right)^{-1}.$$

Proof. Write $\boldsymbol{B} = (\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{V}^{-1}$ and $\boldsymbol{C} = (\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{W} - \boldsymbol{B}$ (dropping subscripts). The asymptotic variance with weight $\boldsymbol{W}$ is $(\boldsymbol{B}+\boldsymbol{C})\boldsymbol{V}(\boldsymbol{B}+\boldsymbol{C})'$. The cross term vanishes because $\boldsymbol{C}\boldsymbol{V}\boldsymbol{B}' = [(\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A} - (\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A}](\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1} = \boldsymbol{0}$. Hence the variance equals $(\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1} + \boldsymbol{C}\boldsymbol{V}\boldsymbol{C}'$, and $\boldsymbol{C}\boldsymbol{V}\boldsymbol{C}'$ is positive semidefinite. □

Intuitively, moments that are measured with more noise get less weight. "Optimal" is relative to a given set of moment conditions: poorly chosen moments remain inefficient even with the best weight. Adding valid moment conditions never increases the asymptotic variance of efficient GMM.

$\boldsymbol{V}_0$ is unknown, so efficient GMM is implemented in steps:

1. Estimate $\hat{\boldsymbol{\theta}}^{(1)}$ with a convenient weight, e.g. $\boldsymbol{W}_n = \boldsymbol{I}$ or, in linear IV, $(\boldsymbol{Z}'\boldsymbol{Z}/n)^{-1}$. It is consistent but inefficient.
2. Compute $\hat{\boldsymbol{V}} = \frac{1}{n}\sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}}^{(1)})\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}}^{(1)})'$ and re-estimate with $\boldsymbol{W}_n = \hat{\boldsymbol{V}}^{-1}$. This **two-step GMM** estimator is asymptotically efficient.

Repeating step 2 until convergence gives **iterated GMM**; letting the weight depend on $\boldsymbol{\theta}$ inside the objective, $Q_n(\boldsymbol{\theta}) = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta})'\hat{\boldsymbol{V}}(\boldsymbol{\theta})^{-1}\bar{\boldsymbol{g}}_n(\boldsymbol{\theta})$, gives the **continuously updated estimator** (CUE). All three are asymptotically equivalent. In finite samples, estimating $\boldsymbol{V}_0$ introduces bias, especially with many moment conditions (Altonji and Segal 1996), so efficient GMM need not beat a simple one-step estimator in practice.

2.6 Testing overidentifying restrictions

GMM consistency rests on $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$. With $q > k$ there is information left to check it: at the efficient estimate the sample moments should be close to zero.

Theorem 2.4 (Hansen's J test (L. P. Hansen 1982))**.** *Let $\hat{\boldsymbol{\theta}}$ be the efficient GMM estimator and $\hat{\boldsymbol{V}}$ a consistent estimator of $\boldsymbol{V}_0$. Under $H_0 : \mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$,*

$$J = n\,\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}})'\,\hat{\boldsymbol{V}}^{-1}\,\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) \;\xrightarrow{d}\; \chi^2_{q-k}.$$

Three remarks. The factor n matters: without it the statistic converges to zero. The degrees of freedom are the number of overidentifying restrictions, $q - k$; in a just-identified model $J = 0$ by construction and nothing can be tested. Finally, the test has power only against violations that cannot be absorbed by $\boldsymbol{\theta}$: if all instruments are invalid in the same direction, J may not reject. A rejection says that *some* moment conditions fail, not which.

2.7 Conditional moments and optimal instruments

Models are often stated as a conditional restriction $\mathbb{E}[\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0) \mid \boldsymbol{z}_i] = 0$, e.g. $\rho = y_i - \boldsymbol{x}_i'\boldsymbol{\beta}$ with exogenous $\boldsymbol{z}_i$. By the law of iterated expectations, *any* function $\boldsymbol{D}(\boldsymbol{z}_i)$ yields valid unconditional moments $\mathbb{E}[\boldsymbol{D}(\boldsymbol{z}_i)\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$, so there are infinitely many GMM estimators. The asymptotically best choice, the **optimal instrument**, is

$$\boldsymbol{D}^*(\boldsymbol{z}_i) = \mathbb{E}\Big[\frac{\partial \rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \,\Big|\, \boldsymbol{z}_i\Big] \Big/ \operatorname{Var}\left(\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0) \mid \boldsymbol{z}_i\right).$$

In the linear model with exogenous regressors ($\boldsymbol{z}_i = \boldsymbol{x}_i$) this is $-\boldsymbol{x}_i/\sigma_i^2$: the efficient moment estimator is weighted least squares (GLS). With endogenous regressors it is $-\mathbb{E}[\boldsymbol{x}_i \mid \boldsymbol{z}_i]/\sigma_i^2$, which motivates 2SLS: the first stage approximates $\mathbb{E}[\boldsymbol{x}_i \mid \boldsymbol{z}_i]$ by a linear projection.

2.8 Linear instrumental variables as GMM

Consider $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + u_i$ with $\boldsymbol{x}_i$ of dimension k and some regressors correlated with u_i (omitted variables, measurement error, simultaneity). OLS is inconsistent. Let $\boldsymbol{z}_i$ be an $l \times 1$ vector of instruments, containing the exogenous regressors, with

- **exogeneity:** $\mathbb{E}[\boldsymbol{z}_i u_i] = \boldsymbol{0}$;
- **relevance:** $\operatorname{rank} \mathbb{E}[\boldsymbol{z}_i \boldsymbol{x}_i'] = k$ (requires $l \geq k$), and $\mathbb{E}[\boldsymbol{z}_i \boldsymbol{z}_i']$ nonsingular.

The moment function is $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\beta}) = \boldsymbol{z}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})$, with sample analogue $\frac{1}{n}\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$ in matrix notation. Minimizing $\frac{1}{n^2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$ gives the closed form

$$\hat{\boldsymbol{\beta}}(\boldsymbol{W}_n) = (\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{y}.$$

Here $\boldsymbol{A}_0 = -\mathbb{E}[\boldsymbol{z}_i \boldsymbol{x}_i']$ and $\boldsymbol{V}_0 = \mathbb{E}[u_i^2 \boldsymbol{z}_i \boldsymbol{z}_i']$. The familiar estimators are special cases.

Table 2.1: Linear GMM estimators

Case	Weight $\boldsymbol{W}_n$	Estimator
Just identified, $l = k$	any	IV: $\hat{\boldsymbol{\beta}}_{IV} = (\boldsymbol{Z}'\boldsymbol{X})^{-1}\boldsymbol{Z}'\boldsymbol{y}$
Overidentified	$(\boldsymbol{Z}'\boldsymbol{Z})^{-1}$	2SLS: $[\boldsymbol{X}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{X}]^{-1}\boldsymbol{X}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{y}$
Overidentified, homoskedastic	$(\hat{\sigma}^2\boldsymbol{Z}'\boldsymbol{Z}/n)^{-1}$ (optimal)	2SLS
Overidentified, heteroskedastic	$(\frac{1}{n}\sum_i \hat{u}_i^2 \boldsymbol{z}_i \boldsymbol{z}_i')^{-1}$ (optimal)	two-step efficient GMM

2SLS is the GMM estimator with $\boldsymbol{W}_n = (\boldsymbol{Z}'\boldsymbol{Z})^{-1}$; under homoskedasticity, $\boldsymbol{V}_0 = \sigma^2\mathbb{E}[\boldsymbol{z}_i\boldsymbol{z}_i']$ and this weight is optimal. Under heteroskedasticity both 2SLS and two-step GMM are consistent, GMM is more efficient, and **2SLS standard errors must be heteroskedasticity robust**:

$$\widehat{\operatorname{Var}}(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\Big(\sum_i \hat{u}_i^2 \boldsymbol{z}_i\boldsymbol{z}_i'\Big)\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X})^{-1}.$$

For two-step GMM this simplifies to $[\boldsymbol{X}'\boldsymbol{Z}(\sum_i \hat{u}_i^2 \boldsymbol{z}_i\boldsymbol{z}_i')^{-1}\boldsymbol{Z}'\boldsymbol{X}]^{-1}$.

Weak instruments. If the instruments are only weakly correlated with the endogenous regressors, $\boldsymbol{A}_0$ is close to rank deficient. IV and 2SLS are then biased towards OLS and their normal approximation is poor, even in large samples (Staiger and Stock 1997). Always report the first-stage regression and the F statistic for the excluded instruments; a common rule of thumb flags $F < 10$ as weak.

2.8.1 Illustration: 2SLS versus efficient GMM

The simulation below has one endogenous regressor, three instruments and errors whose variance depends on the first instrument. Both estimators are consistent, but efficient GMM puts less weight on the noisy moment $\mathbb{E}[z_{1i}u_i] = 0$.

```
set.seed(2027)
one_sample <- function(n = 500) {
  Z <- cbind(1, matrix(rnorm(3 * n), n))              # constant + 3
↪  instruments
  v <- rnorm(n)
```

```
  x <- drop(Z[, 2:4] %*% c(0.4, 0.4, 0.4)) + v
  u <- (0.5 * v + sqrt(0.75) * rnorm(n)) * exp(Z[, 2]) # endogenous,
↪  heteroskedastic
  y <- 1 + x + u
  X <- cbind(1, x)
  gmm <- function(W) solve(t(X) %*% Z %*% W %*% t(Z) %*% X, t(X) %*% Z %*% W
↪  %*% t(Z) %*% y)
  b_2sls <- gmm(solve(crossprod(Z)))                    # step 1: 2SLS
  S      <- crossprod(Z * drop(y - X %*% b_2sls)) / n
  b_gmm  <- gmm(solve(S))                               # step 2: efficient GMM
  e      <- drop(y - X %*% b_gmm)
  gbar   <- colMeans(Z * e)
  J      <- n * drop(t(gbar) %*% solve(crossprod(Z * e) / n) %*% gbar)
  c(b_2sls = b_2sls[2], b_gmm = b_gmm[2], J_reject = J > qchisq(0.95, df = 2))
}
mc <- t(replicate(1000, one_sample()))
round(c(mean_2sls = mean(mc[, 1]), mean_gmm = mean(mc[, 2]),
        sd_2sls = sd(mc[, 1]), sd_gmm = sd(mc[, 2]),
        J_rejection_rate = mean(mc[, 3])), 3)
```

```
#>        mean_2sls         mean_gmm          sd_2sls           sd_gmm
#>            1.016            1.002            0.273            0.199
#> J_rejection_rate
#>            0.039
```

Both estimators are centred on the true slope of one. The standard deviation of efficient GMM is about a quarter smaller than that of 2SLS. The J test, with $q - k = 4 - 2 = 2$ degrees of freedom, rejects the true moment conditions at roughly the nominal 5% rate.

2.9 Two-step estimators as joint GMM

Many estimators proceed in two steps: a first step estimates a nuisance parameter $\boldsymbol{\alpha}$ (a propensity score, a first-stage regression, a selection equation), and a second step estimates the parameter of interest τ given $\hat{\boldsymbol{\alpha}}$. Treating $\hat{\boldsymbol{\alpha}}$ as known in the second step gives wrong standard errors. Stacking both steps into one MM problem gives the correct ones automatically. We first introduce the leading example of this course, the average treatment effect estimated by inverse probability weighting, and then derive the general result.

2.9.1 Background: potential outcomes and the average treatment effect

Potential outcomes. Let $D_i \in \{0, 1\}$ indicate whether unit i receives a treatment (a training programme, smoking during pregnancy, a policy). Each unit has two *potential outcomes*: Y_{1i} if treated and Y_{0i} if not (Rubin 1974). The individual effect is $Y_{1i} - Y_{0i}$. We observe only one of the two,

$$Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i},$$

so individual effects are never observed. We therefore target averages, most often the **average treatment effect**

$$\tau = \mathbb{E}[Y_{1i} - Y_{0i}] = \mu_1 - \mu_0, \qquad \mu_d = \mathbb{E}[Y_{di}],$$

or the average treatment effect on the treated, $\tau_T = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]$. The notation implicitly assumes that a unit's outcome does not depend on the treatment of other units.

Why a simple comparison fails. The difference in observed means decomposes as

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \underbrace{\mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]}_{\tau_T} + \underbrace{\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]}_{\text{selection bias}}.$$

If treated and untreated units differ in their untreated outcomes, e.g. mothers who smoke also differ in age, education and marital status, the comparison is biased. Under random assignment D_i is independent of (Y_{0i}, Y_{1i}), the selection bias is zero and $\tau_T = \tau$.

Selection on observables. Without randomization, suppose we observe covariates $\boldsymbol{x}_i$ that capture all differences between the groups that are relevant for the potential outcomes:

- *Unconfoundedness:* $(Y_{0i}, Y_{1i}) \perp D_i \mid \boldsymbol{x}_i$. Within cells of $\boldsymbol{x}_i$, treatment is as good as random.
- *Overlap:* $0 < p(\boldsymbol{x}) < 1$ for all $\boldsymbol{x}$ in the support, where $p(\boldsymbol{x}_i) = P(D_i = 1 \mid \boldsymbol{x}_i)$ is the **propensity score**.

Unconfoundedness is an identifying assumption and cannot be tested with the data; overlap can be checked by comparing the estimated propensity scores of treated and controls.

Identification by weighting. Under these two assumptions

$$\mu_1 = \mathbb{E}\Big[\frac{D_i Y_i}{p(\boldsymbol{x}_i)}\Big], \qquad \mu_0 = \mathbb{E}\Big[\frac{(1-D_i)Y_i}{1-p(\boldsymbol{x}_i)}\Big], \qquad \tau = \mathbb{E}\Big[\frac{D_i Y_i}{p(\boldsymbol{x}_i)} - \frac{(1-D_i)Y_i}{1-p(\boldsymbol{x}_i)}\Big]. \tag{2.1}$$

The proof uses $D_i Y_i = D_i Y_{1i}$, the law of iterated expectations and unconfoundedness; it is Problem Set 2, Problem 4(a). The intuition: a treated unit with propensity score 0.2 represents five units in the population, of which only one was treated. Weighting each treated unit by $1/p(\boldsymbol{x}_i)$ and each control by $1/(1-p(\boldsymbol{x}_i))$ creates a pseudo-population in which treatment no longer depends on $\boldsymbol{x}_i$. This idea goes back to survey sampling (Horvitz and Thompson 1952); Rosenbaum and Rubin (1983) showed that the propensity score is the key balancing score.

Estimation. Replacing expectations by sample means gives the **inverse probability weighting (IPW)** estimator

$$\hat{\tau}_{IPW} = \frac{1}{n}\sum_{i=1}^{n}\Big[\frac{D_i Y_i}{\hat{p}(\boldsymbol{x}_i)} - \frac{(1-D_i)Y_i}{1-\hat{p}(\boldsymbol{x}_i)}\Big].$$

In finite samples the weights $D_i/\hat{p}(\boldsymbol{x}_i)$ need not average to one. The **normalized** version divides by the sum of the weights within each group,

$$\hat{\tau}_{norm} = \frac{\sum_i D_i Y_i/\hat{p}(\boldsymbol{x}_i)}{\sum_i D_i/\hat{p}(\boldsymbol{x}_i)} - \frac{\sum_i (1-D_i)Y_i/(1-\hat{p}(\boldsymbol{x}_i))}{\sum_i (1-D_i)/(1-\hat{p}(\boldsymbol{x}_i))},$$

and is usually more stable. With limited overlap some $\hat{p}(\boldsymbol{x}_i)$ are close to 0 or 1, a few observations get very large weights, and both estimators become imprecise. Regression adjustment, $\tau = \mathbb{E}[\mathbb{E}(Y_i \mid D_i = 1, \boldsymbol{x}_i) - \mathbb{E}(Y_i \mid D_i = 0, \boldsymbol{x}_i)]$, and doubly robust estimators that combine both models are the main alternatives; Imbens and Wooldridge (2009) survey them.

Why this is a two-step problem. The propensity score is unknown. In practice it is modelled parametrically, typically as a logit $p(\boldsymbol{x}_i; \boldsymbol{\alpha}) = \Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})$, and $\boldsymbol{\alpha}$ is estimated by ML in a first step. The ML first-order condition is a moment condition (Example 2.1), so both steps are MM problems:

$$\boldsymbol{\psi}_1(\boldsymbol{w}_i, \boldsymbol{\alpha}) = (D_i - \Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha}))\boldsymbol{x}_i, \qquad \psi_2(\boldsymbol{w}_i, \boldsymbol{\alpha}, \tau) = \frac{D_i Y_i}{\Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})} - \frac{(1-D_i)Y_i}{1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})} - \tau.$$

The question is how the estimation error in $\hat{\boldsymbol{\alpha}}$ affects the distribution of $\hat{\tau}$.

2.9.2 Stacking the two steps

Let $\boldsymbol{\psi}_1(\boldsymbol{w}_i, \boldsymbol{\alpha})$ (k_1 conditions) define the first step and $\psi_2(\boldsymbol{w}_i, \boldsymbol{\alpha}, \tau)$ the second. The stacked moment function

$$\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}) = \begin{pmatrix} \boldsymbol{\psi}_1(\boldsymbol{w}_i, \boldsymbol{\alpha}) \\ \psi_2(\boldsymbol{w}_i, \boldsymbol{\alpha}, \tau) \end{pmatrix}, \qquad \boldsymbol{\theta} = (\boldsymbol{\alpha}', \tau)',$$

is just identified, and $\boldsymbol{A}_0$ is block lower triangular:

$$\boldsymbol{A}_0 = \begin{pmatrix} \boldsymbol{A}_{11} & \boldsymbol{0} \\ \boldsymbol{A}_{21} & A_{22} \end{pmatrix}, \quad \boldsymbol{A}_{11} = \mathbb{E}\Big[\frac{\partial \boldsymbol{\psi}_1}{\partial \boldsymbol{\alpha}'}\Big], \ \boldsymbol{A}_{21} = \mathbb{E}\Big[\frac{\partial \psi_2}{\partial \boldsymbol{\alpha}'}\Big], \ A_{22} = \mathbb{E}\Big[\frac{\partial \psi_2}{\partial \tau}\Big].$$

Applying Theorem 2.1 and reading off the last row gives the influence representation

$$\sqrt{n}(\hat{\tau} - \tau_0) = -\frac{1}{A_{22}} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \Big[\psi_{2i} - \boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{\psi}_{1i}\Big] + o_p(1).$$

The term $\boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{\psi}_{1i}$ is the **correction for the estimated first step**. It vanishes only if $\boldsymbol{A}_{21} = \boldsymbol{0}$, i.e. if the second-step moment does not depend on $\boldsymbol{\alpha}$ in expectation. Solving the stacked system jointly gives the same point estimates as the sequential procedure, because the first block does not involve τ; only the standard errors change.

For IPW, $A_{22} = -1$ and $\boldsymbol{A}_{21} \neq \boldsymbol{0}$. A surprising but general feature: when the first step is efficient ML of a correctly specified propensity score model, the correction *reduces* the variance, so using the estimated propensity score is more efficient than using the true one (Hirano, Imbens, and Ridder 2003). The naive standard error that treats $\hat{p}(\boldsymbol{x}_i)$ as known is therefore conservative for IPW, but in other two-step problems it can be too small.

```
set.seed(2027)
one_draw <- function(n = 1000) {
  x  <- rnorm(n)
  p  <- plogis(-0.3 + 0.8 * x)            # true propensity score
  d  <- rbinom(n, 1, p)
  y  <- 1 + 2 * x + d * 1 + rnorm(n)       # true ATE = 1
  ph <- fitted(glm(d ~ x, family = binomial))
  ipw <- function(p) mean(d * y / p - (1 - d) * y / (1 - p))
  c(true_p = ipw(p), estimated_p = ipw(ph))
}
sim <- replicate(2000, one_draw())
round(rbind(mean = rowMeans(sim), sd = apply(sim, 1, sd)), 3)
```

```
#>      true_p estimated_p
#> mean  1.000       1.002
#> sd    0.183       0.086
```

Both versions are centred on the true ATE of one, and the estimator with the *estimated* propensity score has the clearly smaller standard deviation.

Problem Set 2, Problem 4 derives the moment functions and the blocks of $\boldsymbol{A}_0$ for IPW, and Computer Tutorial 2 estimates the model jointly, computes the stacked sandwich variance and compares it with the bootstrap.

2.10 Further applications

GMM is the natural framework whenever theory delivers orthogonality conditions rather than a full likelihood. Two examples from the slides:

- **Dynamic panel data.** In $y_{it} = \gamma y_{i,t-1} + \boldsymbol{x}_{it}'\boldsymbol{\beta} + \eta_i + \varepsilon_{it}$, first differencing removes η_i but makes $\Delta y_{i,t-1}$ correlated with $\Delta\varepsilon_{it}$. Lags $y_{i,t-2}, y_{i,t-3}, \ldots$ are valid instruments, and their number grows with t. Arellano and Bond (1991) use all of them in an efficient GMM estimator.
- **Covariate balancing propensity score.** Imai and Ratkovic (2014) estimate the propensity score by GMM, combining the logit score with moment conditions that require the weighted covariate means to balance between treated and controls. The overidentified version allows a J test of the propensity score model.

2.11 Computer tutorial 2

Learning goals. Estimate a two-step estimator as one joint MM problem with `nleqslv`; compute its sandwich variance by hand and compare with the bootstrap; implement 2SLS, efficient two-step GMM and the J test in matrix form and check them against `AER::ivreg`.

Packages. `nleqslv`, `boot`, `sandwich`, `lmtest`, `AER`. **Data.** `cattaneo2.csv` (Cattaneo 2010), 4642 births; `card1995.csv` (Card 1995), 3613 young men from the NLS 1976 cross-section.

2.11.1 Problem 1: The effect of smoking on birth weight (IPW as joint MM)

This problem uses the results of Problem Set 2, Problem 4.

Variable	Description
`bweight`	infant birth weight (grams)
`mbsmoke`	= 1 if the mother smoked during pregnancy
`mmarried`	= 1 if the mother is married
`mage`	mother's age (years)
`medu`	mother's education (years)
`fbaby`	= 1 if first baby

We want the average treatment effect τ of `mbsmoke` on `bweight`. The propensity score is modelled as a logit,

$$P(\texttt{mbsmoke} = 1 \mid X_i) = \Lambda(\alpha_0 + \alpha_1 \texttt{mmarried} + \alpha_2 \texttt{mage} + \alpha_3 \texttt{mage}^2 + \alpha_4 \texttt{fbaby} + \alpha_5 \texttt{medu}).$$

a) Compute the raw difference in mean birth weight between smokers and non-smokers. Why is it unlikely to be the ATE?

b) Write the stacked moment function for $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \tau)'$ from Problem Set 2 as an R function that returns the $k+1$ sample moments, and solve the system with `nleqslv`. Compare $\hat{\boldsymbol{\alpha}}$ with `glm` and $\hat{\tau}$ with the IPW formula evaluated at the `glm` propensity score. Why do they coincide?

c) Estimate the asymptotic variance of $\hat{\boldsymbol{\theta}}$ with the MM sandwich $\hat{\boldsymbol{A}}^{-1}\hat{\boldsymbol{V}}\hat{\boldsymbol{A}}^{-1\prime}/n$, computing $\hat{\boldsymbol{A}}$ analytically. Report the standard error of $\hat{\tau}$ and compare it with the "naive" standard error that treats the propensity score as known.

d) Compute the ATE with normalized weights (weights that sum to one within each treatment group).

e) Estimate the standard errors of both ATE estimators by the nonparametric bootstrap (500 replications). Remember that each bootstrap replication has to re-estimate the propensity score.

2.11.2 Problem 2: Returns to schooling with college proximity (IV and GMM)

Variable	Description
`lwage76`	log hourly wage, 1976
`ed76`	years of education, 1976
`age76`	age in 1976
`black`, `reg76r`, `smsa76r`	= 1 if black; lives in the South; lives in an urban area (1976)
`nearc4a`, `nearc4b`	= 1 if grew up near a 4-year public / private college

Following Card (1995) and B. E. Hansen (2022, Ch. 12), consider

$$\log w_i = \beta_0 + \beta_1 \texttt{educ}_i + \beta_2 \texttt{exper}_i + \beta_3 \texttt{exper}_i^2/100 + \beta_4 \texttt{black}_i + \beta_5 \texttt{south}_i + \beta_6 \texttt{urban}_i + u_i,$$

with potential experience `exper` = `age` − `educ` − 6. Education is endogenous (ability is omitted), and so is experience, because it is a function of education. The excluded instruments are `nearc4a`, `nearc4b`, `age76` and `age76^2/100`.

a) Construct `exper` and `exper2`, drop observations with missing values and estimate the equation by OLS.

b) Estimate the first stage for `ed76`. Test the joint significance of `nearc4a` and `nearc4b` with a heteroskedasticity-robust F test. Are the instruments strong?

c) Count the moment conditions q and parameters k. Compute 2SLS with the GMM formula and $\boldsymbol{W}_n = (\boldsymbol{Z}'\boldsymbol{Z})^{-1}$, with heteroskedasticity-robust standard errors. Check your results with `AER::ivreg` and `sandwich`.

d) Compute the two-step efficient GMM estimator and its standard errors.

e) Compute Hansen's J statistic and its p-value. What does it test here?

f) Compare the estimated returns to schooling across OLS, 2SLS and GMM.

2.12 Exercises

Problem Set 2 contains the analytical exercises for this chapter:

- **Problem 1**: two MM estimators of the Poisson mean, their asymptotic distributions, and efficient GMM combining both moments.
- **Problem 2**: ML, MM and GMM for the gamma distribution.
- **Problem 3**: linear IV as GMM: IV, 2SLS, the optimal weighting matrix and the J test.
- **Problem 4**: identification of the average treatment effect by inverse probability weighting and its estimation as a joint MM problem. Computer tutorial 2, Problem 1 builds on it.

Part II

Computation and flexible models

3 Simulation-Based Estimation

In preparation

This chapter will be added before the corresponding lectures. Planned content:

- Monte Carlo integration and its error
- Quasi-random (Halton) draws and Gaussian quadrature
- Maximum simulated likelihood and method of simulated moments
- Bootstrap
- Computer tutorial 3

4 Nonparametric Methods

> **In preparation**
>
> This chapter will be added before the corresponding lectures. Planned content:
>
> - Histogram and kernel density estimation; bias, variance and bandwidth choice
> - Nadaraya-Watson and local linear regression
> - Rule-of-thumb and cross-validated bandwidths; confidence bands
> - Curse of dimensionality and semiparametric alternatives
> - Computer tutorial 4

5 Introduction to Statistical Learning

In preparation

This chapter will be added before the corresponding lectures. Planned content:

- Prediction vs. estimation; bias-variance trade-off
- Cross-validation
- Ridge, lasso and elastic net
- Trees and ensembles (overview)
- Lasso for inference and double machine learning (outlook)
- Computer tutorial 5

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A Large-Sample Asymptotics

> **In preparation**
>
> This chapter will be added before the corresponding lectures. Planned content:
>
> - Convergence in probability and distribution; Slutsky and continuous mapping
> - WLLN, CLT; delta method
> - Uniform convergence and consistency of extremum estimators

B A Short R Primer

> **In preparation**
>
> This chapter will be added before the corresponding lectures. Planned content:
>
> - Projects, scripts and working directories
> - Data import, data frames and matrices
> - Matrix algebra and OLS by hand
> - Writing functions; optimisation with optim
> - Computer tutorial 0